
13 Demonstrate Progress with Results

New Description

Outcome	Weight
Explore of IT Areas	◆◆◆◆

.

Outcome	Weight
Context of Research Project	◆◆◆◆

.

Outcome	Weight
Communication of Research Outcomes	◆◆◆◆

.

Outcome	Weight
Validation of Objectives and Methodology	◆◆◆◆

.

Outcome	Weight
Application of Skills and Knowledge	◆◆◆◆

.

Date	Author	Comment
2025/09/16 18:53	N/a Nikhil	Ready to Mark
2025/09/16 20:06	N/a Nikhil	Not Started
2025/09/16 20:06	N/a Nikhil	Ready to Mark
2025/09/17 10:01	Ananda Maiti	Complete

DEAKIN UNIVERSITY

RESEARCH TRAINING AND PROJECT

ONTRACK SUBMISSION

Demonstrate Progress with Results

Submitted By:
N/a NIKHIL
s224234862
2025/09/16 20:06

Tutor:
Ananda MAITI

Outcome	Weight
Explore of IT Areas	◆◆◆◆◆
Context of Research Project	◆◆◆◆◆
Communication of Research Outcomes	◆◆◆◆◆
Validation of Objectives and Methodology	◆◆◆◆◆
Application of Skills and Knowledge	◆◆◆◆◆

September 16, 2025



SIT723 Research Techniques & Applications

SIT792 Minor Thesis

Task 4.1P: Research Progress with Results

Student Name: Nikhil

Student ID: 224234862

Supervisor: Dr. Ananda Maiti

Course: SIT723 / SIT792

Project Title: Detection and Simulation of GPS Spoofing in Mobile Robots Using Sensor Cross-Validation

Submission Date: September 16, 2025

1. Introduction

This weekly progress diary documents the comprehensive research progress for my minor thesis project: *Detection and Simulation of GPS Spoofing in Mobile Robots Using Sensor Cross-Validation*. This document serves as evidence of systematic progress toward developing effective GPS spoofing detection mechanisms for autonomous mobile robots through simulation-based validation and machine learning approaches.

The research addresses critical cybersecurity challenges in autonomous navigation systems by investigating three core research questions: (1) primary vulnerabilities of GPS in autonomous mobile robots to spoofing attacks, (2) effective sensor fusion techniques for real-time spoofing detection, and (3) dynamic GPS Trust Index design for quantifying navigation data reliability.

This progress diary demonstrates continuous advancement through literature review completion, research methodology development, simulation environment setup, and preliminary algorithm implementation within the PX4-Gazebo-ROS ecosystem.

2. Weekly Meeting Log

Meeting Configuration

- **Supervisor:** Dr. Ananda Maiti
- **Meeting Frequency:** Weekly (Wednesdays, 4:00 PM)
- **Mode:** Hybrid (Microsoft Teams)
- **Duration:** 45-60 minutes per meeting

- **Documentation:** Meeting transcripts maintained for reference

Meeting Summary Table:

Date	Topics Discussed	Actions/Outcomes
July 30, 2025	Project initialization, research scope definition, literature review approach	Established weekly meeting schedule, defined low-cost focus for detection systems
August 23, 2025	Literature review progress, methodology selection, simulation setup challenges	Guidance on theoretical contribution focus, rate of spoofing vs. detection analysis
August 30, 2025	Research methodology finalization, Docker containerization, academic timeline	Docker container setup for collaboration, trimester timeline clarification
September 8, 2025	Research methods selection, task completion status, presentation preparation	Final methodology selection confirmation, literature organization completion

3. Progress Diary

Week 1-2 (July 21 - August 4, 2025)

Summary:

Initiated project with comprehensive literature review planning and research question formulation. Established supervision agreement and defined research scope focusing on civilian GPS applications for autonomous mobile robots.

Achievements:

- Completed supervision agreement with Dr. Ananda Maiti
- Formulated three research questions addressing GPS vulnerabilities, sensor fusion detection, and dynamic trust assessment
- Identified research gap in lightweight detection algorithms for resource-constrained platforms
- Established research theme framework linking vulnerabilities, detection techniques, and trust quantification

Evidence:

[**Research Questions Framework Document:**] Comprehensive formulation of RQ1 (GPS vulnerabilities), RQ2 (sensor fusion implementation), and RQ3 (trust index design) with clear motivation and research gaps identified.

Impediments/Issues:

- Limited availability of literature specifically addressing mobile robotic platforms
- Need to distinguish civilian vs. military GPS applications appropriately

Supervisor Feedback:

- Focus on civilian applications to avoid dual-use research concerns
- Emphasize low-cost, resource-constrained platform requirements
- Develop methodology suitable for practical deployment scenarios

Week 3-4 (August 5 - August 18, 2025)

Summary:

Conducted systematic literature analysis across 15+ research papers, developing comprehensive understanding of GPS spoofing attack taxonomies and existing detection methods.

Achievements:

- Completed initial literature review covering 15 key research papers
- Identified five primary research methodologies: machine learning detection, sensor fusion, signal analysis, behavioral analysis, and cryptographic authentication
- Developed literature theme association framework linking research questions to existing work
- Created research narrative structure for integrated detection framework

Evidence:

[Literature Review Summary:] Comprehensive analysis of GPS spoofing detection methods including SVM-based classification (98.7% accuracy).

Impediments/Issues:

- Insufficient real-time validation data in reviewed literature
- Limited performance benchmarks for resource-constrained platforms

Supervisor Feedback:

- Strengthen analytical comparison of methodologies
- Focus on practical implementation feasibility
- Develop clear contribution positioning within literature landscape

Week 5-6 (August 19 - September 1, 2025)

Summary:

Advanced to research methodology selection and detailed literature review completion. Selected three complementary approaches: LSTM networks, hybrid sensor fusion with DTW, and multi-layered signal/behavioral analysis.

Achievements:

- Finalized selection of three research methods with detailed justification
- Completed comprehensive literature review (42-page document) with systematic analysis
- Developed comparative methodology analysis framework

- Created research timeline spanning simulation development through hardware validation

Evidence:

[Research Methodology Document:] Complete 10-page methodology framework including LSTM architecture design, sensor fusion algorithms, implementation stack architecture, and ethical considerations for dual-use research management.

Impediments/Issues:

- Complexity of integrating multiple detection approaches
- Timeline constraints for comprehensive validation

Supervisor Feedback:

- Excellent methodology selection with strong theoretical foundation
- Ensure practical implementation focus remains priority
- Develop concrete evaluation metrics for detection performance

Week 7-8 (September 2 - September 15, 2025)**Summary:**

Focused on simulation environment preparation and technical implementation planning. Advanced understanding of PX4 SITL ecosystem and MAVROS integration for spoofing scenario development.

Achievements:

- Established PX4 SITL simulation environment with Gazebo integration
- Developed Docker containerization approach for reproducible research
- Created preliminary GPS spoofing simulation framework
- Prepared comprehensive research methodology documentation for submission

Evidence:

[Simulation Environment Setup:] PX4 SITL running in Docker container with MAVLink communication established. GPS sensor parameters accessible for spoofing simulation with binary spoofing state detection capability identified.

Impediments/Issues:

- Docker container configuration complexity for cross-platform compatibility
- Limited granularity in existing GPS spoofing simulation parameters

Supervisor Feedback:

- Simulation approach provides excellent foundation for algorithm development
- Focus on quantifiable spoofing intensity metrics beyond binary detection

- Prepare demonstration video for presentation showcasing environment capabilities

4. Key Evidence of Achievements

Major Milestone Evidence

Milestone 1: Literature Review Completion

[Comprehensive Literature Analysis]

Evidence: 42-page literature review document analyzing 15+ research papers across GPS spoofing detection methodologies. Systematic comparison of machine learning approaches (SVM 98.7% accuracy).
Impact: Established theoretical foundation and identified clear research gaps in lightweight algorithms for mobile robotic platforms.

Milestone 2: Research Methodology Framework

[Integrated Detection Framework Architecture]

Evidence: Complete methodology document featuring LSTM network architecture for temporal pattern analysis, hybrid sensor fusion with Dynamic Time Warping, and multi-layered signal analysis. Implementation stack designed for simulation-to-hardware pipeline.
Impact: Provides systematic approach to GPS spoofing detection suitable for autonomous mobile robots.

Milestone 3: Simulation Environment Establishment

[PX4-Gazebo-ROS Integration Platform]

Evidence: Functional simulation environment with PX4 SITL, Gazebo physics simulation, and MAVROS communication bridge. Docker containerization enables reproducible research deployment across platforms.
Impact: Establishes controlled environment for systematic algorithm development and validation.

Milestone 4: Research Questions and Scope Definition

[Focused Research Direction]

Evidence: Three well-defined research questions addressing GPS vulnerabilities (RQ1), sensor fusion implementation (RQ2), and dynamic trust assessment (RQ3). Clear alignment with literature themes and practical implementation requirements.
Impact: Provides structured approach to addressing critical gaps in mobile robot navigation security.

5. Implementation Progress and Technical Achievements

Simulation Platform Development

The research has successfully established a comprehensive simulation environment using PX4 SITL with Gazebo integration. This platform enables controlled GPS spoofing scenario generation for systematic algorithm testing. Docker containerization ensures reproducible deployment

across different development environments.

Algorithm Framework Design

Three complementary detection approaches have been architected:

- **LSTM Networks:** Temporal pattern recognition for gradual drift attack detection
- **Sensor Fusion + DTW:** Cross-validation between GPS, IMU, and odometry sensors
- **Multi-layered Analysis:** Signal characteristics combined with behavioral pattern assessment

Research Methodology Validation

The selected methodologies demonstrate strong theoretical foundation with practical implementation feasibility. Comparative analysis shows superior performance potential compared to existing binary detection approaches, with focus on quantifiable trust assessment rather than simple spoofed/authentic classification.

6. Reflection on Research Progress

Summary of Achievements

This trimester has established a solid foundation for GPS spoofing detection research through comprehensive literature analysis, methodology selection, and simulation environment preparation. The work demonstrates clear progression from problem identification through theoretical framework development to practical implementation planning.

Milestones Completed

- Comprehensive literature review with systematic methodology analysis
- Three focused research questions with clear theoretical positioning
- Research methodology framework with detailed implementation architecture
- PX4-Gazebo simulation environment established and containerized
- Preliminary algorithm framework designed for integrated detection approach

Outstanding Work

The next phase will focus on algorithm implementation and validation:

- LSTM model development and training on simulated spoofing scenarios
- Sensor fusion algorithm implementation with Dynamic Time Warping
- Dynamic Trust Index model development and calibration
- Comprehensive performance evaluation across diverse attack scenarios
- Hardware validation preparation for real-world deployment testing

Major Feedback Incorporated

Supervisor feedback has been systematically integrated throughout the research development:

- Focused on civilian applications to address dual-use research concerns
- Emphasized low-cost, resource-constrained platform requirements
- Developed quantifiable metrics for spoofing intensity and detection performance
- Established reproducible research methodology through containerization
- Positioned contributions within broader cybersecurity research landscape

Research Impact and Contributions

This research addresses critical gaps in GPS spoofing detection for autonomous mobile robots through:

- Novel integration of temporal analysis, sensor fusion, and trust assessment
- Practical focus on resource-constrained platform deployment
- Systematic simulation-to-hardware validation methodology
- Open-source contribution framework for reproducible research

The established foundation positions the research to make significant contributions to autonomous system security while maintaining practical applicability for real-world deployment scenarios.

Nikhil, Student ID: 224234862

Date: September 16, 2025